

Phase I: Offline Ingestion, Substrate Storage, & Addressing (Once per Block)

Context Block X_i
 B tokens

X_i

Pointer Generator $\mathcal{G}_\phi$
 Pooling: $\mathcal{O}(Bd)$ | Attention: $\mathcal{O}(B^2d)$

Pointer P_i (k tok)

Pointer Registry $\mathcal{I}$
 Entries: $\{(P_i, M_i)\}_{i=1}^N$
 $P_i \in \mathcal{P}$ (k tokens, $k \ll B$)
 Cached vectors $\mathbf{e}(P_i) \in \mathbb{R}^d$

$\mathcal{S}(X_i) \rightarrow M_i$

Memory Substrate $\mathcal{M}$

- $\mathcal{M}_{\text{text}}$: Raw text storage
- $\mathcal{M}_{\text{kv}}$: Paged KV states
- $\mathcal{M}_{\text{latent}}$: Bottleneck $\mathbf{Z}_i$

Handle M_i

Phase II: Online Query-Conditioned Selection, Dereferencing, & Inference (Abridged)

Incoming Query Q_t

Length $L_Q \rightarrow \text{embed } \mathbf{e}(Q_t)$

$\mathbf{e}(Q_t)$

Resolution Operator $\mathcal{R}_\theta$

Dereference M_j for $j \in J^*$

$$\hat{X}_j = \mathcal{R}_\theta(P_j, M_j, Q_t)$$

Resolved volume: mB_{res} tokens

$\hat{X}_{J^*}$

Active Prompt C_t

- Compact $P_{i \notin J^*}$ ($(N - m)k$ tok)
- Payloads $\hat{X}_{J^*}$ (mB_{res} tok)
- Active query Q_t (L_Q tok)

$P_{i \notin J^*}$

Query Selection Index

$$\text{Score}(P_i, Q_t) = \frac{\langle \mathbf{e}(P_i), \mathbf{e}(Q_t) \rangle}{\|\mathbf{e}(P_i)\| \|\mathbf{e}(Q_t)\|}$$

Top- m Selection: $J^* \subseteq \{1, \dots, N\}$
 Overhead: $\mathcal{C}_{\text{select}} = \mathcal{O}(L_Q d + Nd)$

Forward pass

LLM Backbone f_Θ

Autoregressive Decode $\rightarrow$ Output $\hat{Y}_t$

Selection indices $J^* \subseteq \{1, \dots, N\}$ ($m \ll N$)

Dereference M_j ($j \in J^*$)